

Nirmal Francis Xavier

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EDUCATION

Georgia Institute of Technology

Dec 2026

Master of Science in Computer Science (GPA: 4.0/4.0)

Atlanta

Coursework: ML, Deep Learning, Design of Algorithms, Data & Visual Analytics, Knowledge based AI, Big Data Systems

Indian Institute of Technology Kharagpur

May 2019

Master of Technology in Industrial Engineering (GPA: 9.57/10)

India

Coursework: Operations Research, Statistical Decision Modelling, Systems Modelling and Analysis, Safety Analytics

SKILLS

Programming & Tools: Python, R, SQL, PowerBI, Git, Excel, C++, Pyspark, Javascript, AWS, Docker, DataBricks

Core Competencies: Machine Learning, Optimization, Markov Chain Modelling, Monte Carlo Simulation, Hypothesis Testing, Data Engineering, Deep Learning

EXPERIENCE

RELEX SOLUTIONS | Intern - Supply Chain Analytics

May 2026 - Aug 2026

- Implemented a DC level pooled Bayesian Regression forecasting model for the largest auto spares retailer in North America. Enhanced WAPE by 5% and reduced model over bias by 8% compared to traditional time series models.
- Analyzed inventory and order proposal data to identify and quantify the leading causes of Spoilage and Stockouts for a membership-based warehouse club in Central America.

PROTIUM FINANCE | AVP Data Science

Dec 2022 - Jul 2025

- Built XGBoost binary classifier on AWS SageMaker to predict 18-month repayment risk; reduced 8,000+ features to 13 using Optuna + feature selection (F-score, IV, SHAP), achieving a test AUC of 0.698; used the score to drive lending policy.
- Developed a monthly repayment risk score using Logistic Regression (test ROC AUC: 0.876), with WOE-binned features for robustness and VIF-based feature reduction to minimize multicollinearity; operationalized the score in a prioritization strategy that cut operating costs by 15% and improved on-time repayment outcomes by 5%.
- Built Markov-chain loss forecasting model using roll-rate transitions to project lifetime NPA by segment; used the forecasts to flag high-risk cohorts and guide policy tightening, reducing overall NPA by ~12%.
- Implemented rule-based fuzzy matching algorithm combining SBERT + FAISS retrieval with string-similarity metrics (WRatio, token set, Jaro Winkler) to automate customer verification, reducing manual underwriting by 3000 hours annually.

CUSHMAN & WAKEFIELD | Senior Consultant - Data Analytics

Jan 2022 - Dec 2022

- Built an end-to-end maintenance staffing optimizer for HDB (Singapore): facility-location model to place standby units, discrete event simulation (SimPy) for optimal no. of technicians by site and shift from demand and service times, and a roster optimization model (PuLP) with labor + coverage constraints to cut operational costs by \$1M/year.

GENERAL MILLS | Senior Engineer - Manufacturing Analytics

Oct 2019 - Jan 2022

- Developed real-time defect prediction model using Random Forest for rare “cheese oiling-out” failures using time-lagged process signals. Model optimized for failure capture with F0.5 and delivered ~\$1M annual savings by preventing failures.
- Built mixed-integer programming model in R to optimize changeover sequencing and assign product flow across processing, surge capacity, and packaging lines. Minimized runout waste and achieved ~\$500k/year savings.

PROJECTS

Personalized Online Grocery Out-of-Stock Recommendation Engine (GitHub): Built a product substitute recommendation system. Developed a Gaussian Mixture Model to learn soft product families and generate high-quality substitute candidates, then ranked them with an XGBoost model. Awarded the best project for CS7641 ML Fall'25.

CERTIFICATION (Links)

SQL (Advanced), Hacker Rank, Feb 2024 | Supply Chain Digitization and Management, IIM Mumbai, Mar 2021